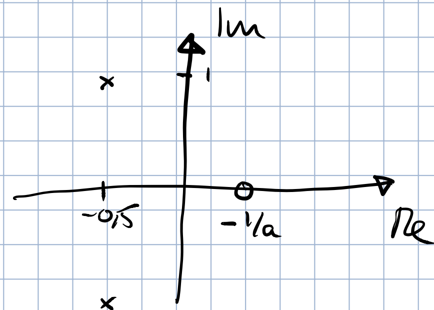
Extra: Nollställe och stegsvar

$$G(s) = \frac{1 + as}{s^2 + s + 1}$$



Poler: $-\frac{1}{2} \pm i\frac{\sqrt{3}}{2}$

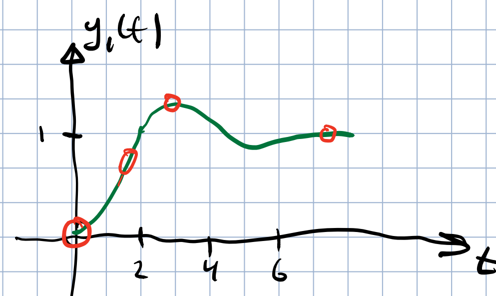
Nollställe: $-1/a$

Stegsvar

$$Y(s) = G(s)\frac{1}{s}$$

$$= \frac{1+as}{s^2+s+1}\frac{1}{s} = \underbrace{\frac{1}{s^2+s+1}\frac{1}{s}}_{Y_1(s)} + a\underbrace{\frac{s}{s^2+s+1}\frac{1}{s}}_{Y_2(s)}$$

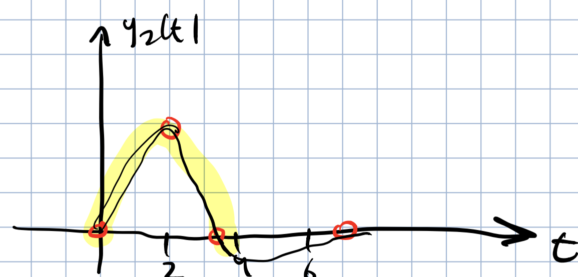
$$y_1(t) = \mathcal{L}^{-1}(Y_1(s))$$



$$y_2(t) = \mathcal{L}^{-1}(Y_2(s))$$

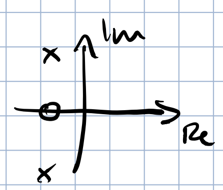
$$= \mathcal{L}^{-1}(sY_1(s)) = \dot{y}_1(t)$$

$$= \frac{d}{dt}y_1(t)$$

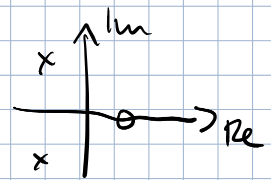


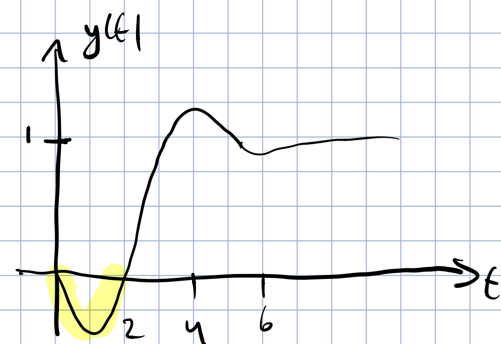
$y(t) = y_1(t) + a\, y_2(t)$ (Totala stegsvaret)

$\boxed{a>0}$



$\boxed{a<0}$





<u>Sammanfattning</u>

- Nollställe påverkar initiala fasen av stegsvaret.
- $a > 0$: Knuff i "rätt" riktning
- $a < 0$: Knuff i "fel" riktning
- Se Föreläsning 7.